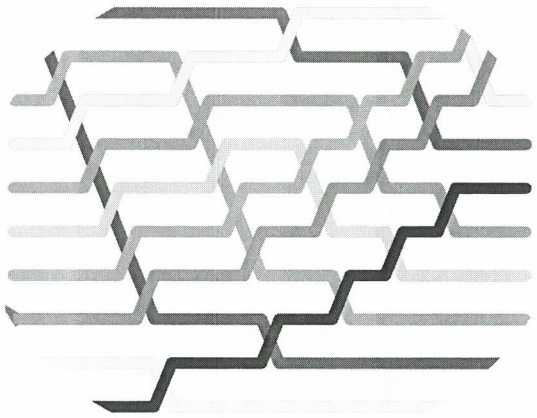
EXERCISE 2 – SORTING ALGORITHMS (41 MARKS)

This is a simple program that provides two functions that, when given an integer list, return the list sorted into ascending order. The first function uses a bubble sort algorithm to sort the given list, while the second function uses a merge sort algorithm.

A program designed to test these functions is shown below (and is provided electronically). Study the code and try to understand what is happening in the program, before attempting the questions that follow.

```
1  def bubbleSort(sortList):
2      sorted = False
3      length = len(sortList)
4      while !sorted:
5          for i in range(length - 2):
6              if sortList[i] > sortList[i+1]:
7                  sortList[i] = sortList[i+1]
8                  sortList[i+1] = sortList[i]
9                  sorted = False
10     return sortList
11
12 def mergeSort(sortList):
13     mid = len(sortList) // 2
14     leftHalf = sortList[:mid]
15     rightHalf = sortList[mid:]
16     if len(sortList) > 1:
18         mergeSort(leftHalf)
19         mergeSort(rightHalf)
20
21 numList = []
22 for i in range(6):
23     print("Add an integer number to the list: ")
24     numList.append(int(input()))
25 print("Bubble sort given:")
26 print(numList)
27 print("Bubble sort returns")
28 print(bubbleSort(numList))
29 input()
```



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SECTION A

A 1 Give a line number from the program that contains a parameter. **[1]**

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A 2 Give a line number from the program that contains recursion. **[1]**

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A 3 Define 'recursion'. **[1]**

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A 4 The program does not run and produces a syntax error.
Explain the cause of this syntax error. **[1]**

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A 5 When the `bubbleSort` function is called, the program gets stuck in an infinite loop.
Explain the cause of this logic error. **[1]**

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A 6 Instead of swapping elements that are in the wrong order, the `bubbleSort` function copies the value of the second element in the pair to the first element in the pair.
State the type of error this is, and explain the cause of the error. **[2]**

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A 7 Currently, the program crashes if the user enters a non-integer value when prompted to add a number to the list. This could be prevented by implementing exception handling.
Explain what exception handling is and why it is necessary. **[2]**

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A 8 The merge sort algorithm is an example of a divide-and-conquer algorithm.
Explain what a 'divide-and-conquer' algorithm is. **[2]**

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A 9 State the time complexity of the bubble sort and merge sort algorithms using Big O notation. **[2]**

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A 10 As well as taking different amounts of time, different algorithms also require different amounts of space in memory. Explain which of the two sorting algorithms has the larger space complexity. **[2]**

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Section A: /15

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